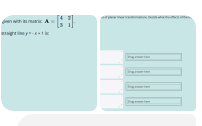


give me answers step by step

Thought for 8s ›

Question 1: Image of line under transformation

Matrix:

$$A = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}$$

So transformation is:

$$T(x,y) = (X,Y)$$
$$X = 4x + 2y$$
$$Y = 3x + y$$

Original line:

$$y = -x + 1$$

Rewrite:

$$x + y = 1$$

Now take two easy points from this line.

Point 1

Let $x = 0$

$$y = 1$$

Point:

$$(0,1)$$

Transform it:

$$X = 4(0) + 2(1) = 2$$
$$Y = 3(0) + 1 = 1$$

So:

$$(0,1) \to (2,1)$$

Point 2

Let $x = 1$

$$y = 0$$

Point:

$$(1,0)$$

Transform it:

$$X = 4(1) + 2(0) = 4$$
$$Y = 3(1) + 0 = 3$$

So:

$$(1,0) \to (4,3)$$

Now find line through transformed points:

$$(2,1),\ (4,3)$$

Slope:

$$m = \frac{3-1}{4-2} = \frac{2}{2} = 1$$

Line form:

$$Y = X + c$$

Use point $(2,1)$:

$$1 = 2 + c$$
$$c = -1$$

So image line is:

$$Y = X - 1$$

Final answer:

$$\boxed{y = x - 1}$$

So select:

✅ $y = x - 1$

Question 2: Effects of matrices on planar figures

Main concept:

For matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

determinant is:

$$\det(A) = ad - bc$$

Rules:

$$|\det(A)|$$

tells area scale.

If determinant is positive, orientation does **not** change.

If determinant is negative, orientation **changes/reverses**.

If determinant is zero, figure collapses into a line.

1st matrix

$$\begin{bmatrix} 5 & 1 \\ 2 & 1 \end{bmatrix}$$
$$\det = 5(1) - 1(2)$$
$$\det = 5 - 2 = 3$$

Area scale:

$$|3| = 3$$

Positive determinant, so orientation does not change.

✅ Answer:

$$\boxed{\text{Areas of figures will triple, orientation will not change.}}$$

2nd matrix

$$\begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}$$
$$\det = 4(2) - 3(3)$$
$$\det = 8 - 9 = -1$$

Area scale:

$$|-1| = 1$$

So area does not change.

Negative determinant, so orientation changes.

✅ Answer:

$$\boxed{\text{Areas of figures do not change, orientation changes.}}$$

3rd matrix

$$\begin{bmatrix} 3 & 3 \\ 3 & 2 \end{bmatrix}$$
$$\det = 3(2) - 3(3)$$
$$\det = 6 - 9 = -3$$

Area scale:

$$|-3| = 3$$

So areas triple.

Negative determinant, so orientation changes.

✅ Answer:

$$\boxed{\text{Areas of figures will triple, orientation changes.}}$$

4th matrix

$$\begin{bmatrix} 3 & 3 \\ 1 & 1 \end{bmatrix}$$
$$\det = 3(1) - 3(1)$$
$$\det = 3 - 3 = 0$$

Determinant zero means transformation is not full 2D anymore.

Figures collapse into line segments.

✅ Answer:

$$\boxed{\text{Two-dimensional figures collapse into sections of straight lines.}}$$

Final Answers

1. $\begin{bmatrix} 5 & 1 \\ 2 & 1 \end{bmatrix}$

✅ **Areas of figures will triple, orientation will not change.**

2. $\begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}$

✅ **Areas of figures do not change, orientation changes.**

3. $\begin{bmatrix} 3 & 3 \\ 3 & 2 \end{bmatrix}$

✅ **Areas of figures will triple, orientation changes.**

4. $\begin{bmatrix} 3 & 3 \\ 1 & 1 \end{bmatrix}$

✅ **Two-dimensional figures collapse into sections of straight lines.**

---

in first question what if i multiply the matrix by AX and get the result where in X|x and in 2nd row y=-x+1| if we replace and solve for x

Let's break this down step-by-step using your approach of multiplying the matrix by X, where X contains the coordinates of points on the line $y = -x + 1$.

The matrix transformation you gave is:

$$A = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}$$

The original line equation is:

$$y = -x + 1 \quad \text{or equivalently} \quad x + y = 1$$

We will represent this equation as a vector of coordinates (x, y), and apply the transformation to it.

Step 1: Matrix Multiplication

Let's consider a generic point (x, y) on the line, and multiply it by the matrix A.

We know from the line equation $y = -x + 1$, so we can substitute $y = -x + 1$ into the matrix equation.

Thus, our vector X is:

$$X = \begin{bmatrix} x \\ -x + 1 \end{bmatrix}$$

Now multiply A by X:

$$A \cdot X = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ -x + 1 \end{bmatrix}$$

Step 2: Perform the Multiplication

Now, let's compute the result:

$$\begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ -x + 1 \end{bmatrix} = \begin{bmatrix} 4x + 2(-x + 1) \\ 3x + (-x + 1) \end{bmatrix} = \begin{bmatrix} 4x - 2x + 2 \\ 3x - x + 1 \end{bmatrix} = \begin{bmatrix} 2x + 2 \\ 2x + 1 \end{bmatrix}$$

Thus, after the transformation, the new coordinates of the point (x, y) on the line $y = -x + 1$ become:

$$(X', Y') = \begin{bmatrix} 2x + 2 \\ 2x + 1 \end{bmatrix}$$

Step 3: Solve for the Equation of the Transformed Line

To find the equation of the transformed line, we need to eliminate x from the transformed coordinates.

Let $X' = 2x + 2$ and $Y' = 2x + 1$.

We now solve for x from one of these equations, say $X' = 2x + 2$:

$$X' - 2 = 2x$$
$$x = \frac{X' - 2}{2}$$

Substitute this expression for x into the equation $Y' = 2x + 1$:

$$Y' = 2\left(\frac{X' - 2}{2}\right) + 1$$
$$Y' = (X' - 2) + 1$$
$$Y' = X' - 1$$

Thus, the equation of the transformed line is:

$$Y' = X' - 1$$

Step 4: Conclusion

The transformed image of the line $y = -x + 1$ under the matrix transformation is:

$$y = x - 1$$

This confirms the same result we got before using the direct approach with points. So, the final transformed line is:

$$\boxed{y = x - 1}$$

---

please tell me step by step how to solve eignvalues and eigenvectors as far as i remember we minus lambda from the diagnols and get determinant and tahn when we get lambda values than we use it in

Let's break down the process of solving eigenvalues and eigenvectors step by step. You are correct that you need to subtract λ from the diagonal elements, but let's go through it thoroughly and step-by-step.

Step 1: Given Matrix

Let's say you have a square matrix A, where you need to find the eigenvalues and eigenvectors. The general form of A is:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Step 2: Eigenvalue Equation

The key equation that governs the eigenvalues λ is:

$$\det(A - \lambda I) = 0$$

Where:

- λ is the eigenvalue,
- I is the identity matrix, and
- det stands for the determinant of the matrix.

The matrix $A - \lambda I$ is obtained by subtracting λ from each diagonal element of the matrix A:

$$A - \lambda I = \begin{bmatrix} a - \lambda & b \\ c & d - \lambda \end{bmatrix}$$

Step 3: Determinant of $A - \lambda I$

Now calculate the determinant of the matrix $A - \lambda I$. For a 2x2 matrix, the determinant is calculated as:

$$\det(A - \lambda I) = (a - \lambda)(d - \lambda) - bc$$

So the equation becomes:

$$\det(A - \lambda I) = (a - \lambda)(d - \lambda) - bc = 0$$

Step 4: Solve for Eigenvalues λ

This equation will be a quadratic equation in λ. Expand it and set it equal to zero:

$$(a - \lambda)(d - \lambda) - bc = 0$$
$$ad - a\lambda - d\lambda + \lambda^2 - bc = 0$$

Rearrange the terms:

$$\lambda^2 - (a + d)\lambda + (ad - bc) = 0$$

This is a standard quadratic equation in λ:

$$\lambda^2 - (a + d)\lambda + (ad - bc) = 0$$

Solve this quadratic equation using the quadratic formula:

$$\lambda = \frac{-(a + d) \pm \sqrt{(a + d)^2 - 4(ad - bc)}}{2}$$

Here, $(a + d)$ is the trace of the matrix, and $(ad - bc)$ is the determinant of the matrix A.

Step 5: Eigenvectors

Once you find the eigenvalues λ_1 and λ_2, you can find the corresponding eigenvectors by solving the following equation:

$$(A - \lambda I)\mathbf{v} = 0$$

Where $\mathbf{v}$ is the eigenvector corresponding to the eigenvalue λ.

To find the eigenvectors, substitute each eigenvalue λ into the equation $A - \lambda I$ and solve the resulting system of linear equations.

For example, if you have $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$, and you find $\lambda = 5$, then you substitute this into $A - 5I$:

$$A - 5I = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}$$

Now, solve the system:

$$\begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This will give you a set of linear equations. You can solve for v_1 and v_2, which will give you the eigenvector corresponding to $\lambda = 5$.

Summary of Steps

1. **Form the matrix equation**: $(A - \lambda I)$
2. **Find the determinant**: $\det(A - \lambda I) = 0$
3. **Solve the quadratic equation** to find the eigenvalues λ.
4. **Substitute each eigenvalue** into $A - \lambda I$ and solve the system to find the corresponding eigenvectors.

Let me know if you want to go through a specific example with numbers!

---

does one eigenvalue yields one eigenvector and tell me what happens if determinant is zero whats that case

Great questions! Let's break it down one by one:

1. Does one eigenvalue yield one eigenvector?

- **No, not necessarily.** For each eigenvalue, there can be **infinitely many eigenvectors** (all scalar multiples of each other). The key idea is that **any scalar multiple** of an eigenvector is still an eigenvector, because it satisfies the eigenvalue equation $A\mathbf{v} = \lambda\mathbf{v}$ as well.
- For example, if $\lambda = 5$ and $\mathbf{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is an eigenvector, then $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$, $\begin{bmatrix} -1 \\ -2 \end{bmatrix}$, or any scalar multiple of $\mathbf{v}$ will also be an eigenvector corresponding to $\lambda = 5$.
- **Eigenvectors are not unique**; they are defined up to scalar multiples. Therefore, for each eigenvalue, there is a **subspace** of eigenvectors corresponding to it, which is often referred to as the **eigenspace**.

2. What happens if the determinant is zero?

When you calculate the **determinant** of $A - \lambda I$ and it turns out to be zero, it means that the matrix A **does not have an inverse**. In the context of eigenvalues, the **determinant being zero** implies a special case:

Case 1: Singular Matrix (Non-Invertible Matrix)

- If the determinant of $A - \lambda I$ equals zero, it means that λ is an eigenvalue and the matrix is **singular** (non-invertible). This occurs when the matrix has a **zero determinant**, which corresponds to the matrix having a **repeated eigenvalue** or **no full rank**.

Case 2: Eigenvalue $\lambda = 0$

- If $\lambda = 0$ is one of the solutions to the characteristic equation (where $\det(A - 0I) = \det(A) = 0$), this indicates that A is **singular** (it does not have an inverse).

Case 3: Defective Matrix

- If you have a repeated eigenvalue (i.e., the determinant is zero and you get the same eigenvalue twice), **the matrix might be defective**. A defective matrix **does not have a full set of linearly independent eigenvectors**.

Example of a Determinant of Zero:

Let's look at a matrix where the determinant is zero.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

The characteristic equation is:

$$\det(A - \lambda I) = \begin{vmatrix} 1 - \lambda & 2 \\ 2 & 4 - \lambda \end{vmatrix} = 0$$

This gives us:

$$(1 - \lambda)(4 - \lambda) - (2)(2) = 0$$
$$(1 - \lambda)(4 - \lambda) - 4 = 0$$
$$4 - \lambda - 4\lambda + \lambda^2 - 4 = 0$$
$$\lambda^2 - 5\lambda = 0$$
$$\lambda(\lambda - 5) = 0$$

So the eigenvalues are $\lambda = 0$ and $\lambda = 5$.

Case 4: For $\lambda = 0$, the matrix is singular:

In this case, the determinant of A is zero, indicating that A is singular (non-invertible), and **the matrix doesn't have full rank**. This means that the matrix compresses the space along some directions, and some dimensions are "collapsed" into lower dimensions.

Summary of What Happens When the Determinant is Zero

- If the determinant of $A - \lambda I$ is zero, it means that λ is an eigenvalue.
- If $\lambda = 0$, the matrix A is singular (non-invertible), and it means that there is **at least one direction** along which the transformation maps vectors to zero (the matrix collapses the space along some axis).
- If the matrix has a **repeated eigenvalue**, it can still have eigenvectors, but the matrix might not have a **full set of linearly independent eigenvectors**. This is called a **defective matrix**.

Example: Non-Invertible (Singular) Matrix with Eigenvalue $\lambda = 0$

Let's consider:

$$A = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$$

Find the determinant of A:

$$\det(A) = 2(2) - 4(1) = 4 - 4 = 0$$

So, the matrix is singular, and it implies that $\lambda = 0$ is an eigenvalue of this matrix. The corresponding eigenvector will give us a vector along which the matrix transformation collapses to zero.

Let me know if you need any further clarification on this!

---

so we dont have to find eigenvectors if determinant is zero

When the determinant of $A - \lambda I$ is **zero**, it means that the matrix A has an eigenvalue λ, and this indicates that the matrix is **singular** (non-invertible). However, **you still need to find the eigenvectors**, even if the determinant is zero. The process doesn't change — the only difference is that you are dealing with a singular matrix, which might have special properties such as having repeated eigenvalues or fewer linearly independent eigenvectors.

Here's the breakdown:

When the Determinant is Zero:

1. **Singular Matrix**: A zero determinant indicates that the matrix A is singular, meaning it doesn't have an inverse. This often happens when the matrix compresses or squashes certain directions in space.
2. **Eigenvalue $\lambda = 0$**: If the determinant is zero, it means that $\lambda = 0$ is one of the eigenvalues. In some cases, λ can be a repeated eigenvalue (i.e., multiple eigenvalues of the same value).
3. **Finding Eigenvectors**: You can still **find eigenvectors corresponding to $\lambda = 0$** or any other eigenvalue by solving the equation:

$$(A - \lambda I)\mathbf{v} = 0$$

Where $\mathbf{v}$ is the eigenvector corresponding to the eigenvalue λ.

- Even though $\lambda = 0$ may indicate a singular matrix, **you still need to solve for eigenvectors** to understand how the matrix behaves when it transforms vectors.

Example: Eigenvalue $\lambda = 0$ (Singular Matrix)

Let's say we have a matrix:

$$A = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$$

Step 1: Find Eigenvalues

We find the eigenvalues by solving $\det(A - \lambda I) = 0$:

$$\det \begin{bmatrix} 2 - \lambda & 4 \\ 1 & 2 - \lambda \end{bmatrix} = 0$$
$$(2 - \lambda)(2 - \lambda) - 4 = 0$$
$$\lambda^2 - 4\lambda = 0$$
$$\lambda(\lambda - 4) = 0$$

So, the eigenvalues are $\lambda = 0$ and $\lambda = 4$.

Step 2: Find Eigenvectors for $\lambda = 0$

Now, for $\lambda = 0$, solve the equation:

$$(A - 0I)\mathbf{v} = 0$$
$$\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives the system of linear equations:

$$2v_1 + 4v_2 = 0$$
$$v_1 + 2v_2 = 0$$

From the second equation:

$$v_1 = -2v_2$$

So the eigenvector corresponding to $\lambda = 0$ is:

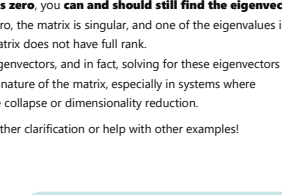
$$\mathbf{v} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

This shows that even if the determinant is zero, you **can still find eigenvectors** corresponding to that eigenvalue.

Conclusion

- **Even if the determinant is zero**, you **can and should still find the eigenvectors**.
- When the determinant is zero, the matrix is singular, and one of the eigenvalues is $\lambda = 0$, which means the matrix does not have full rank.
- The matrix may still have eigenvectors, and in fact, solving for these eigenvectors is essential to understand the nature of the matrix, especially in systems where eigenvalues of zero indicate collapse or dimensionality reduction.

Let me know if you need any further clarification or help with other examples!



explain me this please

Let's walk through the problem step by step and clarify what the transformations mean and whether they are linear transformations or not.

Linear Transformation Definition

A **linear transformation** must satisfy two properties for all vectors $\mathbf{x}, \mathbf{y}$ in $\mathbb{R}^2$, and scalar c:

1. **Additivity**: $\varphi(\mathbf{x} + \mathbf{y}) = \varphi(\mathbf{x}) + \varphi(\mathbf{y})$
2. **Homogeneity (Scalar Multiplication)**: $\varphi(c\mathbf{x}) = c\varphi(\mathbf{x})$

In simple words, the transformation should preserve vector addition and scalar multiplication.

Transformation 1:

$$\varphi(\mathbf{x}) = \begin{bmatrix} x_1 \\ 0 \end{bmatrix}$$

This transformation maps $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ to $\begin{bmatrix} x_1 \\ 0 \end{bmatrix}$.

- **Check Additivity**:
  - $\varphi(\mathbf{x} + \mathbf{y}) = \varphi\left(\begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + y_1 \\ 0 \end{bmatrix}$
  - $\varphi(\mathbf{x}) + \varphi(\mathbf{y}) = \begin{bmatrix} x_1 \\ 0 \end{bmatrix} + \begin{bmatrix} y_1 \\ 0 \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ 0 \end{bmatrix}$
  - They are equal, so the transformation satisfies additivity.
- **Check Homogeneity**:
  - $\varphi(c\mathbf{x}) = \varphi\left(\begin{bmatrix} cx_1 \\ cx_2 \end{bmatrix}\right) = \begin{bmatrix} cx_1 \\ 0 \end{bmatrix}$
  - $c\varphi(\mathbf{x}) = c\begin{bmatrix} x_1 \\ 0 \end{bmatrix} = \begin{bmatrix} cx_1 \\ 0 \end{bmatrix}$
  - They are equal, so the transformation satisfies homogeneity.
- **Conclusion**: This is a **linear transformation** because it satisfies both properties.

Transformation 2:

$$\varphi(\mathbf{x}) = \begin{bmatrix} x_1 + x_2 \\ 0 \end{bmatrix}$$

This transformation maps $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ to $\begin{bmatrix} x_1 + x_2 \\ 0 \end{bmatrix}$.

- **Check Additivity**:
  - $\varphi(\mathbf{x} + \mathbf{y}) = \varphi\left(\begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix}\right) = \begin{bmatrix} (x_1 + y_1) + (x_2 + y_2) \\ 0 \end{bmatrix}$
  - $\varphi(\mathbf{x}) + \varphi(\mathbf{y}) = \begin{bmatrix} x_1 + x_2 \\ 0 \end{bmatrix} + \begin{bmatrix} y_1 + y_2 \\ 0 \end{bmatrix} = \begin{bmatrix} (x_1 + x_2) + (y_1 + y_2) \\ 0 \end{bmatrix}$
  - They are equal, so the transformation satisfies additivity.
- **Check Homogeneity**:
  - $\varphi(c\mathbf{x}) = \varphi\left(\begin{bmatrix} cx_1 \\ cx_2 \end{bmatrix}\right) = \begin{bmatrix} cx_1 + cx_2 \\ 0 \end{bmatrix} = c\begin{bmatrix} x_1 + x_2 \\ 0 \end{bmatrix}$
  - $c\varphi(\mathbf{x}) = c\begin{bmatrix} x_1 + x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} cx_1 + cx_2 \\ 0 \end{bmatrix}$
  - They are equal, so the transformation satisfies homogeneity.
- **Conclusion**: This is a **linear transformation** because it satisfies both properties.

Transformation 3:

$$\varphi(\mathbf{x}) = \begin{bmatrix} 2x_1 \\ 3x_2 \end{bmatrix}$$

This transformation maps $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ to $\begin{bmatrix} 2x_1 \\ 3x_2 \end{bmatrix}$.

- **Check Additivity**:
  - $\varphi(\mathbf{x} + \mathbf{y}) = \varphi\left(\begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix}\right) = \begin{bmatrix} 2(x_1 + y_1) \\ 3(x_2 + y_2) \end{bmatrix} = \begin{bmatrix} 2x_1 + 2y_1 \\ 3x_2 + 3y_2 \end{bmatrix}$
  - $\varphi(\mathbf{x}) + \varphi(\mathbf{y}) = \begin{bmatrix} 2x_1 \\ 3x_2 \end{bmatrix} + \begin{bmatrix} 2y_1 \\ 3y_2 \end{bmatrix} = \begin{bmatrix} 2x_1 + 2y_1 \\ 3x_2 + 3y_2 \end{bmatrix}$
  - They are equal, so the transformation satisfies additivity.
- **Check Homogeneity**:
  - $\varphi(c\mathbf{x}) = \varphi\left(\begin{bmatrix} cx_1 \\ cx_2 \end{bmatrix}\right) = \begin{bmatrix} 2(cx_1) \\ 3(cx_2) \end{bmatrix} = c\begin{bmatrix} 2x_1 \\ 3x_2 \end{bmatrix}$
  - $c\varphi(\mathbf{x}) = c\begin{bmatrix} 2x_1 \\ 3x_2 \end{bmatrix} = \begin{bmatrix} 2(cx_1) \\ 3(cx_2) \end{bmatrix}$
  - They are equal, so the transformation satisfies homogeneity.
- **Conclusion**: This is a **linear transformation** because it satisfies both properties.

Transformation 4:

$$\varphi(\mathbf{x}) = \begin{bmatrix} x_1 x_2 \\ 0 \end{bmatrix}$$

This transformation maps $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ to $\begin{bmatrix} x_1 x_2 \\ 0 \end{bmatrix}$.

- **Check Additivity**:
  - $\varphi(\mathbf{x} + \mathbf{y}) = \varphi\left(\begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix}\right) = \begin{bmatrix} (x_1 + y_1)(x_2 + y_2) \\ 0 \end{bmatrix}$
  - $\varphi(\mathbf{x}) + \varphi(\mathbf{y}) = \begin{bmatrix} x_1 x_2 \\ 0 \end{bmatrix} + \begin{bmatrix} y_1 y_2 \\ 0 \end{bmatrix} = \begin{bmatrix} x_1 x_2 + y_1 y_2 \\ 0 \end{bmatrix}$
  - These are not equal because $(x_1 + y_1)(x_2 + y_2) \neq x_1 x_2 + y_1 y_2$ in general.
- **Conclusion**: This is **not a linear transformation** because it does not satisfy additivity.

Final Answer:

1. $\varphi(\mathbf{x}) = \begin{bmatrix} x_1 \\ 0 \end{bmatrix}$: **a linear transformation** because it is sum- and ratio-preserving.
2. $\varphi(\mathbf{x}) = \begin{bmatrix} x_1 + x_2 \\ 0 \end{bmatrix}$: **a linear transformation** because it is sum- and ratio-preserving.
3. $\varphi(\mathbf{x}) = \begin{bmatrix} 2x_1 \\ 3x_2 \end{bmatrix}$: **a linear transformation** because it is sum- and ratio-preserving.
4. $\varphi(\mathbf{x}) = \begin{bmatrix} x_1 x_2 \\ 0 \end{bmatrix}$: **not a linear transformation** because it is not sum-preserving or ratio-preserving.

The matrix of a spatial transformation is:
M=[[[311686113]]]

i need step by step solution of this matrix with full explanation

Let's go step by step to understand the given matrix M and its properties.

Given Matrix:

$$M = \begin{bmatrix} 3 & 1 & 1 \\ 6 & 8 & 6 \\ 1 & 1 & 3 \end{bmatrix}$$

This matrix represents a **spatial transformation**. To analyze this matrix, we need to explore a few things, such as:

1. **The type of transformation** (whether it's linear, rotation, scaling, etc.)
2. **Eigenvalues and eigenvectors** (which can help us understand the transformation's properties)
3. **Determinant** (to determine if the transformation is invertible)
4. **Trace** (to see if there's any connection to rotation or shear)

Let's break it down:

Step 1: Check if the transformation is linear

A spatial transformation represented by a matrix is always **linear** as long as the matrix is used to transform vectors in space. So, this transformation is **linear**.

Step 2: Compute the Determinant

The **determinant** of a matrix tells us whether the transformation is invertible. If the determinant is **non-zero**, the transformation is invertible. If it's **zero**, the transformation is **singular** (non-invertible).

For the 3x3 matrix M, the determinant is calculated using the following formula:

$$\det(M) = m_{11} \cdot \begin{vmatrix} m_{22} & m_{23} \\ m_{32} & m_{33} \end{vmatrix} - m_{12} \cdot \begin{vmatrix} m_{21} & m_{23} \\ m_{31} & m_{33} \end{vmatrix} + m_{13} \cdot \begin{vmatrix} m_{21} & m_{22} \\ m_{31} & m_{32} \end{vmatrix}$$

Let's calculate this for M:

$$\det(M) = 3 \cdot \begin{vmatrix} 8 & 6 \\ 1 & 3 \end{vmatrix} - 1 \cdot \begin{vmatrix} 6 & 6 \\ 1 & 3 \end{vmatrix} + 1 \cdot \begin{vmatrix} 6 & 8 \\ 1 & 1 \end{vmatrix}$$

Now, let's compute the 2x2 determinants:

1. $\begin{vmatrix} 8 & 6 \\ 1 & 3 \end{vmatrix} = (8 \cdot 3) - (6 \cdot 1) = 24 - 6 = 18$
2. $\begin{vmatrix} 6 & 6 \\ 1 & 3 \end{vmatrix} = (6 \cdot 3) - (6 \cdot 1) = 18 - 6 = 12$
3. $\begin{vmatrix} 6 & 8 \\ 1 & 1 \end{vmatrix} = (6 \cdot 1) - (8 \cdot 1) = 6 - 8 = -2$

Now substitute these values back into the determinant formula:

$$\det(M) = 3 \cdot 18 - 1 \cdot 12 + 1 \cdot (-2)$$
$$\det(M) = 54 - 12 - 2 = 40$$

So, the determinant of M is:

$$\det(M) = 40$$

Step 3: Compute the Trace

The **trace** of a matrix is the sum of its diagonal elements. For matrix M, the trace is:

$$\text{Tr}(M) = 3 + 8 + 3 = 14$$

The trace can give insight into the matrix's scaling properties, and sometimes it is related to the sum of the eigenvalues.

Step 4: Eigenvalues and Eigenvectors

To understand the geometric properties of the transformation, we can calculate the **eigenvalues** and **eigenvectors** of the matrix. This can help identify if the transformation involves rotation, scaling, or shearing.

Eigenvalues:

To find the eigenvalues of the matrix M, we need to solve the **characteristic equation**:

$$\det(M - \lambda I) = 0$$

Where I is the identity matrix, and λ represents the eigenvalue. The equation becomes:

$$M - \lambda I = \begin{bmatrix} 3 - \lambda & 1 & 1 \\ 6 & 8 - \lambda & 6 \\ 1 & 1 & 3 - \lambda \end{bmatrix}$$

Now, compute the determinant of $M - \lambda I$:

$$\det(M - \lambda I) = (3 - \lambda) \cdot \begin{vmatrix} 8 - \lambda & 6 \\ 1 & 3 - \lambda \end{vmatrix} - 1 \cdot \begin{vmatrix} 6 & 6 \\ 1 & 3 - \lambda \end{vmatrix} + 1 \cdot \begin{vmatrix} 6 & 8 - \lambda \\ 1 & 1 \end{vmatrix}$$

Expanding this determinant and solving for λ will give us the eigenvalues. Due to the complexity, this would normally be done using a symbolic math solver or by hand for a more specific case.

Eigenvectors:

Once the eigenvalues are found, we substitute each eigenvalue λ into the equation $(M - \lambda I)\mathbf{v} = 0$ and solve for the corresponding eigenvector $\mathbf{v}$. This is done by row reducing the augmented matrix $[M - \lambda I | 0]$.

Step 5: Interpretation of Results

- **Determinant**: Since the determinant is non-zero ($\det(M) = 40$), the matrix is **invertible**. This means that the transformation is **bijective** (i.e., it does not collapse any part of space to a lower dimension).
- **Trace**: The trace ($\text{Tr}(M) = 14$) can provide some information about the nature of the eigenvalues, but we need to calculate the eigenvalues for a clearer picture.
- **Eigenvalues/Eigenvectors**: These will tell us the scaling, rotation, or shear properties of the transformation. If the eigenvalues are real and positive, it's likely a scaling transformation. If they are complex, it could indicate rotation.

Conclusion:

1. **The matrix represents a linear transformation** because it's a 3x3 matrix, and spatial transformations are generally linear.
2. **The determinant is non-zero (40)**, which means the transformation is **invertible** and there's no collapse in space.
3. **The trace (14)** gives a rough indication of the sum of the eigenvalues, but we need to calculate them for a detailed understanding.
4. **Eigenvalues and eigenvectors** can be computed to understand whether the transformation involves scaling, rotation, or shearing.

Let me know if you want to calculate the eigenvalues and eigenvectors explicitly!

---

det(M−λI)=(3−λ)·
8−λ163−λ
−1·
6163−λ
+1·
618−λ1
give me this determinant step by step

Sure! Let's compute the determinant of $M - \lambda I$ step by step.

Step 1: Matrix Setup

We are given the matrix M:

$$M = \begin{bmatrix} 3 & 1 & 1 \\ 6 & 8 & 6 \\ 1 & 1 & 3 \end{bmatrix}$$

And we need to subtract λI, where I is the identity matrix. This results in:

$$M - \lambda I = \begin{bmatrix} 3 - \lambda & 1 & 1 \\ 6 & 8 - \lambda & 6 \\ 1 & 1 & 3 - \lambda \end{bmatrix}$$

Now we need to find the determinant of this matrix:

$$\det(M - \lambda I) = \begin{vmatrix} 3 - \lambda & 1 & 1 \\ 6 & 8 - \lambda & 6 \\ 1 & 1 & 3 - \lambda \end{vmatrix}$$

Step 2: Expanding the Determinant

The formula for the determinant of a 3x3 matrix is:

$$\det(A) = a_{11} \cdot \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \cdot \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \cdot \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Using the elements of $M - \lambda I$, we expand the determinant as follows:

$$\det(M - \lambda I) = (3 - \lambda) \cdot \begin{vmatrix} 8 - \lambda & 6 \\ 1 & 3 - \lambda \end{vmatrix} - 1 \cdot \begin{vmatrix} 6 & 6 \\ 1 & 3 - \lambda \end{vmatrix} \backslash + 1 \cdot \begin{vmatrix} 6 & 8 - \lambda \\ 1 & 1 \end{vmatrix}$$

Step 3: Compute the 2x2 Determinants

Now we compute the 2x2 determinants one by one:

First Determinant:

$$\begin{vmatrix} 8 - \lambda & 6 \\ 1 & 3 - \lambda \end{vmatrix} = (8 - \lambda)(3 - \lambda) - (6)(1)$$
$$= (8 - \lambda)(3 - \lambda) - 6$$

Expanding the product:

$$= 24 - 8\lambda - 3\lambda + \lambda^2 - 6$$
$$= \lambda^2 - 11\lambda + 18$$

Second Determinant:

$$\begin{vmatrix} 6 & 6 \\ 1 & 3 - \lambda \end{vmatrix} = (6)(3 - \lambda) - (6)(1)$$
$$= 18 - 6\lambda - 6$$
$$= 12 - 6\lambda$$

Third Determinant:

$$\begin{vmatrix} 6 & 8 - \lambda \\ 1 & 1 \end{vmatrix} = (6)(1) - (8 - \lambda)(1)$$
$$= 6 - (8 - \lambda)$$
$$= 6 - 8 + \lambda$$
$$= \lambda - 2$$

Step 4: Substitute Back into the Determinant Formula

Now substitute all these values back into the determinant expansion:

$$\det(M - \lambda I) = (3 - \lambda) \cdot (\lambda^2 - 11\lambda + 18) - 1 \cdot (12 - 6\lambda) + 1 \cdot (\lambda - 2)$$

Simplifying this expression:

$$\det(M - \lambda I) = (3 - \lambda)(\lambda^2 - 11\lambda + 18) - (12 - 6\lambda) + (\lambda - 2)$$

Now, expand $(3 - \lambda)(\lambda^2 - 11\lambda + 18)$:

$$= 3(\lambda^2 - 11\lambda + 18) - \lambda(\lambda^2 - 11\lambda + 18)$$
$$= 3\lambda^2 - 33\lambda + 54 - \lambda^3 + 11\lambda^2 - 18\lambda$$
$$= -\lambda^3 + 14\lambda^2 - 51\lambda + 54$$

Now, substitute the remaining terms:

$$\det(M - \lambda I) = -\lambda^3 + 14\lambda^2 - 51\lambda + 54 - 12 + 6\lambda + \lambda - 2$$

Simplifying:

$$= -\lambda^3 + 14\lambda^2 - 44\lambda + 40$$

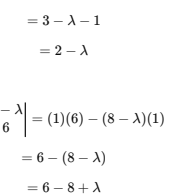
Final Answer

The characteristic polynomial (determinant) of $M - \lambda I$ is:

$$\det(M - \lambda I) = -\lambda^3 + 14\lambda^2 - 44\lambda + 40$$

This is the polynomial equation whose roots are the **eigenvalues** of the matrix M.

Let me know if you would like to continue from here and find the eigenvalues!



The matrix of a spatial transformation is: $\mathbf{M} = \begin{bmatrix} 3 & 6 & 1 \\ 1 & 8 & 1 \\ 1 & 6 & 3 \end{bmatrix}$

give me eigenvalues of this matrix step by step

Let's go step by step to compute the eigenvalues of the given matrix M:

Given Matrix M:

$$M = \begin{bmatrix} 3 & 6 & 1 \\ 1 & 8 & 1 \\ 1 & 6 & 3 \end{bmatrix}$$

Step 1: Eigenvalue Equation

To find the eigenvalues of a matrix, we need to solve the **characteristic equation**:

$$\det(M - \lambda I) = 0$$

Where:

- λ is the eigenvalue,
- I is the identity matrix.

So, $M - \lambda I$ becomes:

$$M - \lambda I = \begin{bmatrix} 3 - \lambda & 6 & 1 \\ 1 & 8 - \lambda & 1 \\ 1 & 6 & 3 - \lambda \end{bmatrix}$$

Step 2: Calculate the Determinant of $M - \lambda I$

Now we need to compute the determinant of the matrix $M - \lambda I$:

$$\det(M - \lambda I) = \begin{vmatrix} 3 - \lambda & 6 & 1 \\ 1 & 8 - \lambda & 1 \\ 1 & 6 & 3 - \lambda \end{vmatrix}$$

We'll expand the determinant using cofactor expansion along the first row:

$$\det(M - \lambda I) = (3 - \lambda) \cdot \begin{vmatrix} 8 - \lambda & 1 \\ 6 & 3 - \lambda \end{vmatrix} - 6 \cdot \begin{vmatrix} 1 & 1 \\ 1 & 3 - \lambda \end{vmatrix} + 1 \cdot \begin{vmatrix} 1 & 8 - \lambda \\ 1 & 6 \end{vmatrix}$$

Now, let's compute each of the 2x2 determinants:

First Determinant:

$$\begin{vmatrix} 8 - \lambda & 1 \\ 6 & 3 - \lambda \end{vmatrix} = (8 - \lambda)(3 - \lambda) - (1)(6)$$
$$= (8 - \lambda)(3 - \lambda) - 6$$

Expanding the product:

$$= 24 - 8\lambda - 3\lambda + \lambda^2 - 6$$
$$= \lambda^2 - 11\lambda + 18$$

Second Determinant:

$$\begin{vmatrix} 1 & 1 \\ 1 & 3 - \lambda \end{vmatrix} = (1)(3 - \lambda) - (1)(1)$$
$$= 3 - \lambda - 1$$
$$= 2 - \lambda$$

Third Determinant:

$$\begin{vmatrix} 1 & 8 - \lambda \\ 1 & 6 \end{vmatrix} = (1)(6) - (8 - \lambda)(1)$$
$$= 6 - (8 - \lambda)$$
$$= 6 - 8 + \lambda$$
$$= \lambda - 2$$

Step 3: Substitute the 2x2 Determinants Back

Now substitute all the values back into the determinant expansion:

$$\det(M - \lambda I) = (3 - \lambda) \cdot (\lambda^2 - 11\lambda + 18) - 6 \cdot (2 - \lambda) + 1 \cdot (\lambda - 2)$$

Step 4: Expand the Equation

First, expand the product $(3 - \lambda) \cdot (\lambda^2 - 11\lambda + 18)$:

$$(3 - \lambda)(\lambda^2 - 11\lambda + 18) = 3(\lambda^2 - 11\lambda + 18) - \lambda(\lambda^2 - 11\lambda + 18)$$
$$= 3\lambda^2 - 33\lambda + 54 - \lambda^3 + 11\lambda^2 - 18\lambda$$
$$= -\lambda^3 + 14\lambda^2 - 51\lambda + 54$$

Now, simplify the second part $-6 \cdot (2 - \lambda)$:

$$-6 \cdot (2 - \lambda) = -12 + 6\lambda$$

Finally, simplify the last term $1 \cdot (\lambda - 2)$:

$$\lambda - 2$$

Step 5: Combine All Terms

Now, we combine all the terms together:

$$\det(M - \lambda I) = -\lambda^3 + 14\lambda^2 - 51\lambda + 54 - 12 + 6\lambda + \lambda - 2$$
$$= -\lambda^3 + 14\lambda^2 - 44\lambda + 40$$

Step 6: Solve the Characteristic Polynomial

So, the characteristic equation is:

$$-\lambda^3 + 14\lambda^2 - 44\lambda + 40 = 0$$

This cubic equation will give us the eigenvalues of the matrix. To find the eigenvalues, we need to solve this cubic equation, which can be done using various methods (e.g., factoring, numerical methods, or using a symbolic math solver).

Would you like to proceed with solving this cubic equation, or would you prefer I help you with the steps to solve it numerically?

---

what are the eigenvalues

To find the eigenvalues of the matrix, we need to solve the **characteristic equation**:

$$-\lambda^3 + 14\lambda^2 - 44\lambda + 40 = 0$$

Step 1: Use Rational Root Theorem

One way to solve the cubic equation is by first checking possible rational roots. According to the **Rational Root Theorem**, the possible rational roots are factors of the constant term (40) divided by factors of the leading coefficient (-1). So the possible rational roots are:

$$\pm 1, \pm 2, \pm 4, \pm 5, \pm 8, \pm 10, \pm 20, \pm 40$$

Step 2: Test Possible Roots

Let's test a few possible values of λ by substituting them into the characteristic equation.

Try $\lambda = 2$:

$$-\lambda^3 + 14\lambda^2 - 44\lambda + 40 = -(2)^3 + 14(2)^2 - 44(2) + 40$$
$$= -8 + 14(4) - 44(2) + 40$$
$$= -8 + 56 - 88 + 40 = 0$$

So, $\lambda = 2$ is a root.

Step 3: Factor the Cubic Polynomial

Since $\lambda = 2$ is a root, we can factor $(\lambda - 2)$ out of the cubic polynomial. We can do this by polynomial division or synthetic division.

Perform Synthetic Division:

Dividing $-\lambda^3 + 14\lambda^2 - 44\lambda + 40$ by $(\lambda - 2)$:

$$\begin{array}{r|rrrr} 2 & -1 & 14 & -44 & 40 \\ & & -2 & 24 & -40 \\ \hline & -1 & 12 & -20 & 0 \end{array}$$

The quotient is:

$$-\lambda^2 + 12\lambda - 20$$

So, we can factor the cubic polynomial as:

$$-\lambda^3 + 14\lambda^2 - 44\lambda + 40 = (\lambda - 2)(-\lambda^2 + 12\lambda - 20)$$

Step 4: Solve the Quadratic

Now, solve the quadratic equation:

$$-\lambda^2 + 12\lambda - 20 = 0$$

Multiply through by -1:

$$\lambda^2 - 12\lambda + 20 = 0$$

Use the **quadratic formula**:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For the equation $\lambda^2 - 12\lambda + 20 = 0$, we have $a = 1$, $b = -12$, and $c = 20$. Substituting these values into the quadratic formula:

$$\lambda = \frac{-(-12) \pm \sqrt{(-12)^2 - 4(1)(20)}}{2(1)}$$
$$\lambda = \frac{12 \pm \sqrt{144 - 80}}{2}$$
$$\lambda = \frac{12 \pm \sqrt{64}}{2}$$
$$\lambda = \frac{12 \pm 8}{2}$$

So, the two roots are:

$$\lambda = \frac{12 + 8}{2} = 10 \quad \text{or} \quad \lambda = \frac{12 - 8}{2} = 2$$

Step 5: Conclusion

The eigenvalues of the matrix are:

$$\boxed{\lambda_1 = 2, \lambda_2 = 10, \lambda_3 = 2}$$

Thus, the eigenvalues are 2, 10, and 2 (with 2 being a repeated eigenvalue).

---

what does reperated eigen values mean like what does it mean

Repeated eigenvalues refer to the case where an eigenvalue appears **more than once** in the set of eigenvalues for a given matrix. This is also known as a **degenerate eigenvalue** or **algebraic multiplicity** greater than 1.

Let's break this down:

Algebraic Multiplicity:

- **Algebraic multiplicity** refers to the number of times a particular eigenvalue appears as a root of the characteristic equation. For example, if an eigenvalue $\lambda = 2$ appears twice in the list of eigenvalues, then its algebraic multiplicity is 2.
- In your case, we found the eigenvalues $\lambda = 2, 2, 10$. Here, $\lambda = 2$ has an **algebraic multiplicity** of 2, meaning it is repeated twice, while $\lambda = 10$ appears only once.

Geometric Multiplicity:

- **Geometric multiplicity** refers to the number of linearly independent eigenvectors associated with an eigenvalue. It is the dimension of the eigenspace corresponding to that eigenvalue.
- **Geometric multiplicity** can never be greater than the **algebraic multiplicity**. If the geometric multiplicity is equal to the algebraic multiplicity, the matrix is said to be **diagonalizable**. If the geometric multiplicity is smaller, the matrix is not diagonalizable and is called **defective**.

What does repeated eigenvalues mean?

- **Repeated eigenvalues** indicate that the transformation described by the matrix has some special properties. Here's a bit more about what this could mean:
  - **Scaling behavior**: A repeated eigenvalue can indicate that the matrix scales vectors along certain directions multiple times.
  - **Symmetry**: In many cases, repeated eigenvalues are a sign of symmetry or special properties of the transformation.
  - **Defective matrices**: If the matrix has repeated eigenvalues but doesn't have enough linearly independent eigenvectors, the matrix is **defective**. This means the matrix cannot be diagonalized, and we can't fully simplify the transformation to a scaling operation along different axes.
  - **Diagonalizability**: If the geometric multiplicity matches the algebraic multiplicity, it means we can find enough linearly independent eigenvectors to diagonalize the matrix.

Example with your matrix:

For your matrix M, we found the eigenvalue $\lambda = 2$ repeated twice. This tells us a couple of things:

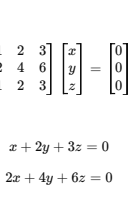
1. **Eigenvalue $\lambda = 2$** corresponds to at least two linearly independent directions in space (i.e., two eigenvectors). This could mean that the transformation affects two directions in the same way.
2. **Eigenvalue $\lambda = 10$** corresponds to a single direction in space.

If you have two eigenvectors corresponding to $\lambda = 2$ (with geometric multiplicity 2), the matrix will be diagonalizable. Otherwise, the matrix might be **defective** and might not have enough eigenvectors to form a complete basis of eigenvectors for $\mathbb{R}^3$.

Key Points to Remember:

- Repeated eigenvalues can indicate symmetry or special transformations.
- The **algebraic multiplicity** counts how many times an eigenvalue appears.
- The **geometric multiplicity** counts how many linearly independent eigenvectors exist for that eigenvalue.
- If the geometric multiplicity is less than the algebraic multiplicity, the matrix is **defective** and **not diagonalizable**.

Let me know if you need further clarification!



solve this

Thought for 7s ›

Given matrix:

$$C = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix}$$

We need to find:

1. **Kernel / Null space**
2. **Image / Column space**

Step 1: Check the rows

Rows are:

$$R_1 = [1, 2, 3]$$
$$R_2 = [2, 4, 6]$$
$$R_3 = [1, 2, 3]$$

Notice:

$$R_2 = 2R_1$$
$$R_3 = R_1$$

So only **one independent row** exists.

Therefore:

$$\text{rank}(C) = 1$$

Step 2: Find kernel space

Kernel means:

$$C \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

So:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives equations:

$$x + 2y + 3z = 0$$
$$2x + 4y + 6z = 0$$
$$x + 2y + 3z = 0$$

But second and third equations are just copies of the first one.

So the real condition is only:

$$x + 2y + 3z = 0$$

This is a **plane**, not a line.

Since there is 1 equation in 3 variables:

$$\dim(\ker C) = 3 - 1 = 2$$

So kernel is **2-dimensional**.

But the correct plane is:

$$x + 2y + 3z = 0$$

Not:

$$2x - y = 0$$

So the kernel option given with plane $2x - y = 0$ is wrong.

Step 3: Find image space

Image space means column space.

Columns are:

$$C_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$
$$C_2 = \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$
$$C_3 = \begin{bmatrix} 3 \\ 6 \\ 3 \end{bmatrix} = 3\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

All columns are multiples of:

$$\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

So image space is:

$$\text{Im}(C) = \text{span}\left\{\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}\right\}$$

That means image space is **1-dimensional**.

Parametric form:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = t\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

So:

$$x = t, \quad y = 2t, \quad z = t$$

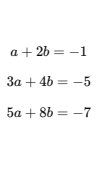
Correct Statement

✅ **The image space is 1-dimensional; it contains vectors parallel to the line**

$$x = t, \quad y = 2t, \quad z = t$$

So select only this one:

The image space is 1-dimensional; it contains vectors parallel to the line $x = t$, $y = 2$



in the part B of this question u know i found the image space as 2 and nullspace or kernel space 1 and i also know the equation of kernel space i.e x=3t y=-t, z=t but i dont know how to find the equation of image space

Let's work through part **(b)** of the problem step by step to find the **image space** (column space) of the matrix B.

Given Matrix B:

$$B = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 4 & -5 \\ 5 & 8 & -7 \end{bmatrix}$$

Step 1: Image Space (Column Space)

The **image space** is spanned by the columns of the matrix B. It's the set of all possible linear combinations of the columns of the matrix.

So, we start by writing down the **columns** of B:

$$\mathbf{c}_1 = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}, \quad \mathbf{c}_2 = \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}, \quad \mathbf{c}_3 = \begin{bmatrix} -1 \\ -5 \\ -7 \end{bmatrix}$$

To find the image space, we need to determine **if any of these columns are linearly dependent**. If some columns are linearly dependent, they do not add new directions to the image space, and we would reduce the dimension of the image space.

Step 2: Check for Linear Dependence

To check for linear dependence, we need to see if any column is a scalar multiple of another. A quick check shows:

- Notice that:

$$\mathbf{c}_2 = 2\mathbf{c}_1$$
$$\begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

- Now, check if $\mathbf{c}_3$ is a linear combination of $\mathbf{c}_1$ and $\mathbf{c}_2$. To do this, we set up a system of equations to find scalars a and b such that:

$$a\mathbf{c}_1 + b\mathbf{c}_2 = \mathbf{c}_3$$

Substitute the vectors:

$$a\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} + b\begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix} = \begin{bmatrix} -1 \\ -5 \\ -7 \end{bmatrix}$$

This gives the system:

$$a + 2b = -1$$
$$3a + 4b = -5$$
$$5a + 8b = -7$$

From the first two equations:

1. $a + 2b = -1$
2. $3a + 4b = -5$

Solve equation (1) for a:

$$a = -1 - 2b$$

Substitute this into equation (2):

$$3(-1 - 2b) + 4b = -5$$
$$-3 - 6b + 4b = -5$$
$$-3 - 2b = -5$$
$$-2b = -2 \quad \Rightarrow \quad b = 1$$

Substitute $b = 1$ into $a = -1 - 2b$:

$$a = -1 - 2(1) = -3$$

So, we find:

$$a = -3, \quad b = 1$$

Thus:

$$\mathbf{c}_3 = -3\mathbf{c}_1 + 1\mathbf{c}_2$$

This shows that $\mathbf{c}_3$ is **linearly dependent** on $\mathbf{c}_1$ and $\mathbf{c}_2$.

Step 3: Image Space (Column Space)

Since $\mathbf{c}_3$ is a linear combination of $\mathbf{c}_1$ and $\mathbf{c}_2$, the image space of the matrix is spanned by just $\mathbf{c}_1$ and $\mathbf{c}_2$.

Thus, the image space is **2-dimensional**, and it is spanned by the two independent columns $\mathbf{c}_1$ and $\mathbf{c}_2$.

Step 4: Equation of the Image Space

The image space (column space) is spanned by the vectors:

$$\text{Im}(B) = \text{span}\left\{\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}\right\}$$

This means any vector in the image space can be written as a linear combination of $\mathbf{c}_1$ and $\mathbf{c}_2$:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = a\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} + b\begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}$$

Where a and b are scalars.

Thus, the equation of the image space is the set of all possible vectors formed by linear combinations of $\mathbf{c}_1$ and $\mathbf{c}_2$.

Conclusion

- The **image space** (column space) is **2-dimensional**.
- The image space is spanned by the vectors $\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}$.

This means the image space contains all possible linear combinations of these two vectors.

Let me know if you'd like further clarification on any steps!

---

image space) is
2x + y − z = 0. how to found this equation

To find the **equation of the image space**, we're looking for a way to represent the set of all possible vectors in the image space of the matrix B.

Given Matrix B:

$$B = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 4 & -5 \\ 5 & 8 & -7 \end{bmatrix}$$

We previously found that the image space is spanned by the first two columns, which are:

$$\mathbf{c}_1 = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}, \quad \mathbf{c}_2 = \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}$$

This means the image space is the set of all linear combinations of these two vectors:

$$\text{Im}(B) = \text{span}\left\{\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}\right\}$$

So, any vector in the image space can be written as:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = a\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} + b\begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}$$

Where a and b are scalar coefficients.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a + 2b \\ 3a + 4b \\ 5a + 8b \end{bmatrix}$$

Now, we want to find the equation that describes the image space. This will be the relationship between x, y, and z that holds for any vector in the image space.

Step 1: Set up a system of equations

The first component gives:

$$x = a + 2b$$

The second component gives:

$$y = 3a + 4b$$

The third component gives:

$$z = 5a + 8b$$

We now have a system of three equations in terms of two variables a and b. To find the equation that describes the image space, we eliminate a and b.

Step 2: Eliminate a and b

We can eliminate a and b by solving for a and b in terms of x, y, and z.

From the equation for x, we have:

$$x = a + 2b \quad \Rightarrow \quad a = x - 2b$$

Now, substitute this expression for a into the equations for y and z.

For y:

$$y = 3a + 4b = 3(x - 2b) + 4b = 3x - 6b + 4b = 3x - 2b$$

So, we get:

$$y = 3x - 2b$$

Now solve for b:

$$2b = 3x - y \quad \Rightarrow \quad b = \frac{3x - y}{2}$$

Now substitute this expression for b into the equation for z:

$$z = 5a + 8b = 5(x - 2b) + 8b = 5x - 10b + 8b = 5x - 2b$$

Substitute $b = \frac{3x - y}{2}$ into this equation:

$$z = 5x - 2 \cdot \frac{3x - y}{2}$$
$$z = 5x - (3x - y)$$
$$z = 5x - 3x + y$$
$$z = 2x + y$$

Step 3: Final equation of the image space

The final equation that describes the image space is:

$$\boxed{2x + y - z = 0}$$

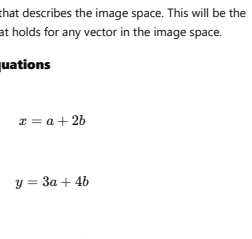
This equation describes all vectors in the image space of matrix B, which is the set of all possible linear combinations of the two columns $\mathbf{c}_1$ and $\mathbf{c}_2$.

Conclusion:

- The **image space** is a plane in $\mathbb{R}^3$, and the equation of this plane is:

$$2x + y - z = 0$$

This is the relationship between the components of the vectors in the image space.



Consider the following transformations of $\mathbb{R}^2 \to \mathbb{R}^2$. Decide whether they are linear transformations or not.

- $\varphi(\mathbf{x}) = \begin{bmatrix} x_1^2 \\ 0 \end{bmatrix}$: This is not a linear transformation, because it is not sum preserving or not ratio preserving.
- $\varphi(\mathbf{x}) = \begin{bmatrix} x_1 + x_2 \\ 0 \end{bmatrix}$: This is a linear transformation, because it is sum- and ratio-preserving.
- $\varphi(\mathbf{x}) = \begin{bmatrix} 2x_1 \\ 3x_2 \end{bmatrix}$: This is a linear transformation, because it is sum- and ratio-preserving.
- $\varphi(\mathbf{x}) = \begin{bmatrix} x_1 x_2 \\ 0 \end{bmatrix}$: This is not a linear transformation, because it is not sum preserving or not ratio preserving.

give me more practice questions like this

Here are some more practice questions similar to the one you uploaded, focusing on determining whether a transformation is linear or not:

Practice Questions

1. Consider the following transformations of $\mathbb{R}^2 \to \mathbb{R}^2$. Decide whether they are linear transformations or not.

(a) $\varphi(\mathbf{x}) = \begin{bmatrix} x_1^2 \\ x_2 \end{bmatrix}$

- This is ___________ because it is ____________.

(b) $\varphi(\mathbf{x}) = \begin{bmatrix} x_1 + x_2 \\ x_1 - x_2 \end{bmatrix}$

- This is ___________ because it is ____________.

(c) $\varphi(\mathbf{x}) = \begin{bmatrix} 3x_1 \\ 2x_2 \end{bmatrix}$

- This is ___________ because it is ____________.

(d) $\varphi(\mathbf{x}) = \begin{bmatrix} x_1 x_2 \\ 0 \end{bmatrix}$

- This is ___________ because it is ____________.

2. Consider the following transformations of $\mathbb{R}^2 \to \mathbb{R}^2$. Determine if they are linear transformations.

(a) $\varphi(\mathbf{x}) = \begin{bmatrix} x_1^2 + 3x_2 \\ x_1 + 4x_2 \end{bmatrix}$

- This is ___________ because it is ____________.

(b) $\varphi(\mathbf{x}) = \begin{bmatrix} 2x_1 + x_2 \\ x_1 - x_2 \end{bmatrix}$

- This is ___________ because it is ____________.

(c) $\varphi(\mathbf{x}) = \begin{bmatrix} x_1 + 2 \\ x_2 + 5 \end{bmatrix}$

- This is ___________ because it is ____________.

(d) $\varphi(\mathbf{x}) = \begin{bmatrix} x_1 x_2 \\ x_1 + x_2 \end{bmatrix}$

- This is ___________ because it is ____________.

3. Determine whether the following transformations are linear or not.

(a) $\varphi(\mathbf{x}) = \begin{bmatrix} x_1 + 2x_2 \\ x_1^2 - x_2 \end{bmatrix}$

- This is ___________ because it is ____________.

(b) $\varphi(\mathbf{x}) = \begin{bmatrix} x_1 - 2x_2 \\ 4x_1 + 3x_2 \end{bmatrix}$

- This is ___________ because it is ____________.

(c) $\varphi(\mathbf{x}) = \begin{bmatrix} x_1 x_2 \\ x_1 + x_2 + 1 \end{bmatrix}$

- This is ___________ because it is ____________.

(d) $\varphi(\mathbf{x}) = \begin{bmatrix} 3x_1 + x_2 \\ x_2 - x_1 \end{bmatrix}$

- This is ___________ because it is ____________.

Answer Key Template:

- **Linearity Check**:
  - **Sum-preserving**: Check if $\varphi(\mathbf{x} + \mathbf{y}) = \varphi(\mathbf{x}) + \varphi(\mathbf{y})$
  - **Scalar multiplication preserving**: Check if $\varphi(c\mathbf{x}) = c\varphi(\mathbf{x})$
- **Non-linearity** usually comes from:
  - Non-linear terms (like x_1^2, $x_1 x_2$, etc.)
  - Constant terms that break the additivity or homogeneity rules.

Would you like me to solve a few of these for you as examples? Let me know!